

Warm Up:

1. $4 + 3 + 2 \cdot 5$

→ $4 + 3 + 10 = 17$

2. $2/5 - 1/10 \div 3/5$

→ $1/10 \div 3/5 = 1/6$; $2/5 - 1/6 = 12/30 - 5/30 = 7/30$

3. $(25 - 4^2) \cdot 3 + 2$

→ $(25 - 16)(3) + 2 = 27 + 2 = 29$

4. $(16 - 3^2) + (2 + 3)^2$

→ $7 + 25 = 32$

5. Ravi earned \$30 on Monday and twice as much on Tuesday. On Wednesday, he spent half of the money he earned on Monday and Tuesday. How much did he have left?

→ Mon 30, Tue 60 → 90 total; spent $90 \div 2 = 45$; he has \$45 left

1.3 Homework Questions?

Speak Like A Mathematician:

coefficient: The numerical factor of a term containing a variable.

like terms: Terms with the same variable raised to the same power.

simplest form: An expression with no like terms left to combine and no parentheses.

equivalent expressions: Expressions that have the same value for every value of the variable.

Key Concept — The Distributive Property

Distributive Property	
Words:	To multiply a sum or difference by a number, multiply EACH term inside by the number outside.
Symbols:	$a(b + c) = ab + ac$ and $a(b - c) = ab - ac$
Examples:	$4(3 + 5) = 4(3) + 4(5) = 12 + 20 = 32$

A look ahead...

Using the Symmetric Property, you can write the distributive property in reverse order:

$$\rightarrow ab + ac = a(b + c)$$

Careful: the printed line reads $ab + bc = a(b + c)$, which is not true – $ab + bc$ factors to $b(a + c)$. The correct statement is $ab + ac = a(b + c)$.

This is called factoring.

Example 1: Applying the Distributive Property

- In a recent year, the median salary for an engineer in the United States was \$55,000 and the median salary for a computer programmer was \$52,000.
 - Write an expression to estimate the cost for a business to employ an engineer and a computer programmer for 5 years.
$$\rightarrow 5(55,000 + 52,000) \text{ or } 5(55,000) + 5(52,000)$$
 - Evaluate the expression to estimate the total cost for a business to hire both for 5 years.
$$\rightarrow 5(107,000) = \$535,000$$

Example 2: Mental Math

- $4(23)$
 $\rightarrow 4(20 + 3) = 80 + 12 = 92$
- $5 \cdot 99$
 $\rightarrow 5(100 - 1) = 500 - 5 = 495$
- $4 \cdot 1002$
 $\rightarrow 4(1000 + 2) = 4000 + 8 = 4008$
- $5(4 \frac{1}{5})$
 $\rightarrow 5(4 + \frac{1}{5}) = 20 + 1 = 21$

Example 3: Distribute an Algebraic Expression

- $4(5x - 7)$
 $\rightarrow 20x - 28$
- $(3y^2 + y - 8)6$
 $\rightarrow 18y^2 + 6y - 48$
- $-6(r + 3g - t)$
 $\rightarrow -6r - 18g + 6t$ – distribute the **NEGATIVE** to every term
- $(-2m + 5b - 2k)2$
 $\rightarrow -4m + 10b - 4k$
- $-4(4a + 2b - (\frac{1}{2})c)$
 $\rightarrow -16a - 8b + 2c$

Combine Like Terms (CONCEPT CHECK)

Simplify each expression, if possible.

- $14a + 18a$
 $\rightarrow 32a$
- $4b^2 + 9b - 3b$
 $\rightarrow 4b^2 + 6b$ – b^2 and b are **NOT** like terms
- $b^2 + 13b + 13$
 $\rightarrow$ Already in simplest form – no like terms to combine.

Example 4: Write and Simplify Expressions

- three times the sum of $2x$ and $3y$, decreased by twice the difference of 4 and y
 $\rightarrow 3(2x + 3y) - 2(4 - y) = 6x + 9y - 8 + 2y = 6x + 11y - 8$
- the product of 9 and t squared, increased by 3 times the sum of 2 and t squared
 $\rightarrow 9t^2 + 3(2 + t^2) = 9t^2 + 6 + 3t^2 = 12t^2 + 6$

Example 5: Simplify Using the Distributive Property

- $6d + 4(3d + 5)$
 $\rightarrow 6d + 12d + 20 = 18d + 20$
- $4(fg + 3g) + 5g$
 $\rightarrow 4fg + 12g + 5g = 4fg + 17g$ – fg and g are not like terms
- $2(n + 2n - 5)$
 $\rightarrow 2(3n - 5) = 6n - 10$ (or distribute first: $2n + 4n - 10 = 6n - 10$)

Tonight's Homework: Paper Practice 1.4 (#1-12)